Quick Start Guide

Emotion Analysis from App Reviews - Replication Package

This artifact provides essential resources to replicate the study on fine-grained emotion extraction from mobile app reviews, presented at the 33rd IEEE International Requirements Engineering Conference 2025.

Key Components:

- Dataset: App reviews annotated with emotions through human annotation (ground truth).
- Annotation guidelines: For human and LLM-based annotation of app reviews with an emotion taxonomy.
- Code: For automated emotion annotation using GPT-4o, Mistral Large 2, and Gemini 2.0 Flash models.

Quick Setup:

1. Clone the artifact: https://doi.org/10.6084/m9.figshare.28548638
2. Set up Python environment:
   a. `python -m venv venv`
   b. `source venv/bin/activate` (Linux/Mac) or `source venv/Scripts/activate` (Windows)
3. Install dependencies:
   a. `pip install -r requirements.txt`
4. Add your LLM API key to '.env' file.
5. Run annotation scripts:
   a. Create assistant:
      i. `python code/llm_annotation/create_assistant_openai.py --model gpt-4o`
   b. Annotate reviews:
      i. `python code/llm_annotation/annotate_emotions_openai.py --input data/ground-truth.xlsx`

Artifact License: GPL-3.0 License. Refer to LICENSE file for details.

Citation: Q. Motger, M. Oriol, M. Tiessler, X. Franch, and J. Marco, "What About Emotions? Guiding Fine-Grained Emotion Extraction from Mobile App Reviews," in Proc. IEEE Int. Requirements Eng. Conf. (RE), 2025.

For detailed guidance, please refer to the provided README file.